LECTURE 2
AC AND DC CHARGERS

Charging is an essential process for maintaining the performance, reliability, and service life of electric vehicle (EV) batteries. Improper charging can damage a battery and cause premature failure. For example, leaving a lead-acid battery at a low state of charge for a long period can result in sulphation, which reduces its performance and capacity. Similarly, excessive charging voltage, current, or temperature can damage battery cells. Therefore, modern EV batteries require careful control of charging current and voltage rather than simply connecting them to a constant-voltage or constant-current source.

Most EV charging systems obtain electrical energy from the AC power grid. Renewable energy sources, such as solar photovoltaic panels, can also be used as part of the charging infrastructure. Since the battery stores energy in DC form, AC electricity must be converted into DC before it can charge the battery. The charger must also regulate the output according to the battery's voltage, current, temperature, and charging requirements. The two main types of conductive charging systems are AC charging and DC charging, which differ mainly in the location of the power-conversion equipment.

CLASSIFICATION OF EV CHARGERS

EV chargers can be classified based on energy source, charger installation, power flow, charging level, connector type, and supply phase.

1**. Based on energy source**: EV chargers are classified into AC chargers and DC chargers. AC chargers supply alternating current to the vehicle, where the onboard charger converts it into DC for battery charging. DC chargers convert AC into regulated DC outside the vehicle and supply it directly to the battery.

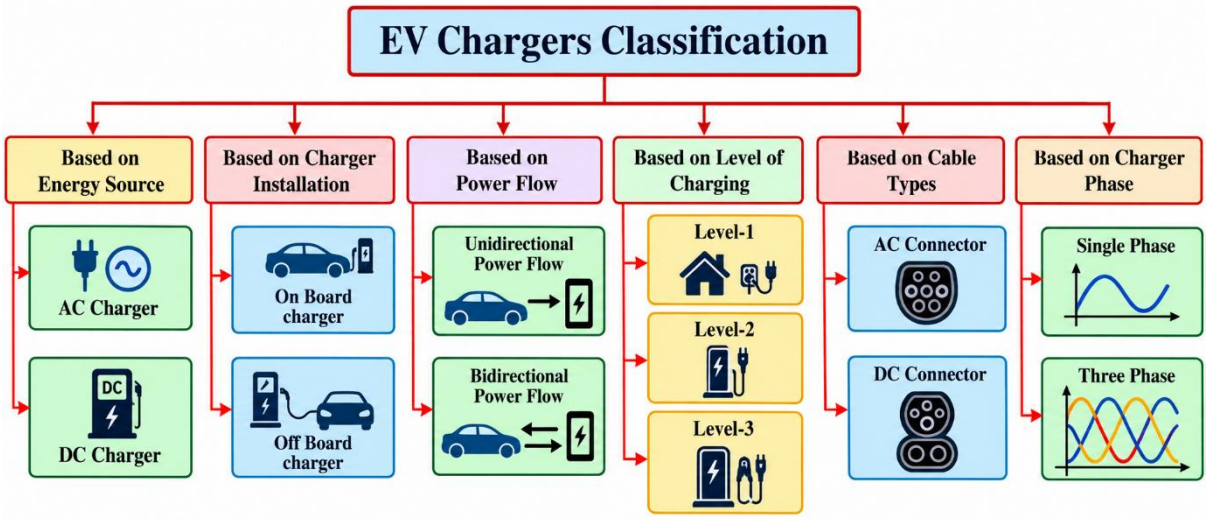
2. **Based on charger installation**: Chargers are classified as onboard chargers and offboard chargers. An onboard charger is installed inside the vehicle and generally has a limited power rating due to space and weight constraints. An offboard charger is installed outside the vehicle, usually at a charging station, and can support higher charging power.

3. **Based on power flow**: Chargers are classified as unidirectional and bidirectional. A unidirectional charger transfers energy in one direction, from the grid to the vehicle battery. A bidirectional charger allows energy to flow in both directions, enabling the battery to supply power back to the grid or compatible loads, as in vehicle-to-grid (V2G) and vehicle-to-home (V2H) applications.

4. **Based on level of charging**: Chargers are classified as Level 1, Level 2, and Level 3. Level 1 charging generally uses a standard, low-power AC supply and is suitable for long charging periods. Level 2 provides higher AC charging power and reduces charging time. Level 3 generally refers to high-power charging in the classification shown; it commonly uses offboard DC fast charging. The exact ratings depend on the applicable standard.

5. **Based on cable and connector type**: Chargers use AC connectors or DC connectors. AC connectors supply AC power to the vehicle's onboard charger, whereas DC connectors deliver regulated DC power from an offboard charger to the battery system. Connector designs vary according to regional standards and charging requirements.

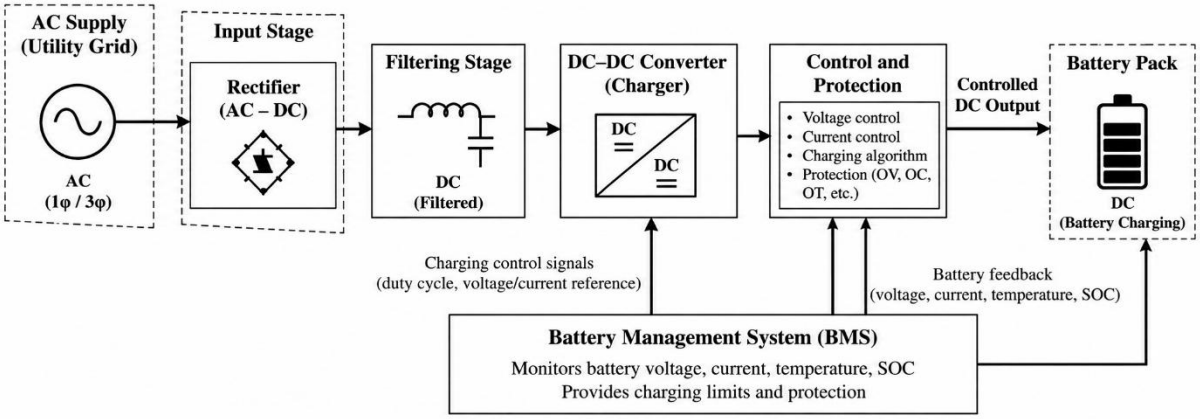
6. **Based on charger supply phase**: Chargers are classified as single-phase or three-phase. Single-phase chargers use a single-phase AC supply and are commonly used for residential charging. Three-phase chargers use a three-phase supply and can support higher power levels, depending on the charger and vehicle capabilities.



BASIC PRINCIPLE OF BATTERY CHARGING

In a conventional grid-connected charging system, electrical energy is supplied as alternating current (AC), whereas the battery requires direct current (DC). Therefore, the AC supply must pass through a rectification stage to convert it into DC. The resulting DC voltage must be sufficiently smooth, with very little ripple, to ensure proper charging.

Ripple is the unwanted variation present in a rectified DC output. If the voltage falls below the battery voltage during part of a cycle, charging may temporarily stop. If the voltage rises excessively, it can damage the battery. As charging current increases, producing a smooth and well-regulated DC output becomes more demanding. Consequently, high-power chargers require appropriately designed rectification, filtering, power-conversion, and control circuits.



A modern battery charger regulates both current and voltage to meet the battery's requirements. The charger and the battery management system (BMS) work together to maintain safe operating conditions. The BMS monitors cell voltages, current, and temperature and communicates charging limits or protective commands as required.

AC CHARGING SYSTEMS

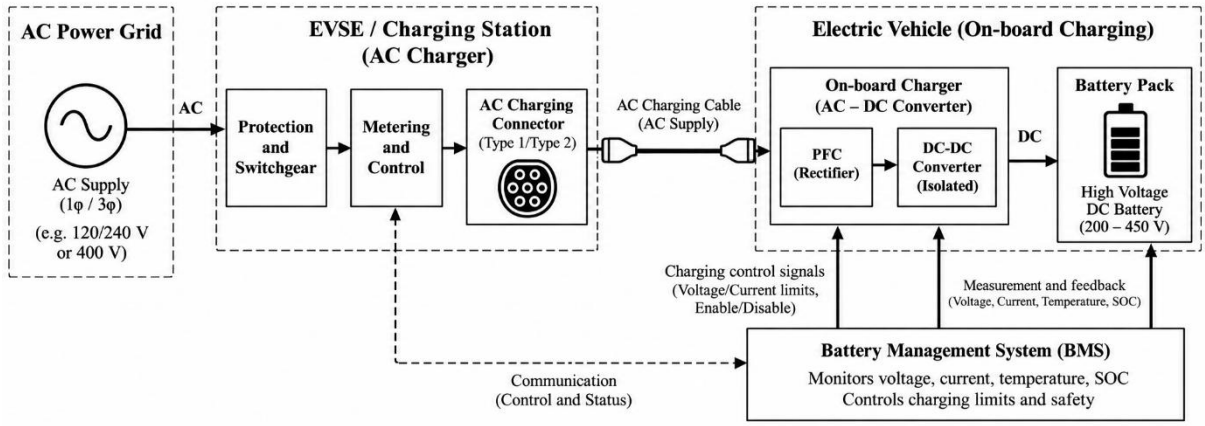
In an AC charging system, alternating current from the electrical grid is supplied to the vehicle through Electric Vehicle Supply Equipment (EVSE). The vehicle's onboard charger (OBC) converts the incoming AC into DC suitable for charging the battery. The onboard charger generally includes an AC–DC converter, a power factor correction (PFC) circuit, and charging-control circuits.

The PFC circuit improves the power factor and helps reduce unwanted current harmonics drawn from the electrical supply. The charging-control circuit regulates the power delivered to the battery according to its requirements and coordinates with the BMS and vehicle control system.

Since the onboard charger is carried inside the vehicle, it must meet automotive requirements for reliability, vibration resistance, thermal performance, and service life. It also adds weight, occupies space, and contributes to vehicle cost. Its power rating limits the maximum AC charging rate.

AC Charging Architecture

In an AC charging architecture, the power grid supplies AC electricity to the EV through EVSE. The AC supply enters the onboard charger, where it is converted into DC by an AC–DC converter. The onboard charger may include a PFC circuit to improve the power factor and reduce current harmonics. The regulated DC output is supplied to the battery under the supervision of the BMS.



AC Level 1 Charging

AC Level 1 charging uses a standard household electrical supply. It operates at approximately 120 V AC, single-phase, with typical current ratings of 12 A or 16 A. These correspond to power levels of approximately 1.08 kW and 1.44 kW, respectively.

Because the available power is relatively low, Level 1 charging takes longer than higher-power charging methods. It is suitable when the vehicle remains connected for an extended period, such as overnight. The AC supply enters the vehicle, and the onboard charger converts it into DC for the battery.

Table 1: Charge method electrical ratings according to SAE EV AC charging power levels.

Charge Method	Nominal Supply Voltage	Maximum Current	Branch Circuit Breaker Rating	Output Power Level
AC Level 1	120 V AC, 1-phase	12 A	15 A	1080 W
AC Level 1	120 V AC, 1-phase	16 A	20 A	1440 W
AC Level 2	208–240 V AC, 1-phase	16 A	20 A	3300 W
AC Level 2	208–240 V AC, 1-phase	32 A	40 A	6600 W
AC Level 2	208–240 V AC, 1-phase	≤80 A	Per NEC 635	≤14.4 kW

AC Level 2 Charging

AC Level 2 charging uses a higher supply voltage, typically 208–240 V AC, single-phase. The current rating depends on the charging equipment and may be 16 A, 32 A, or higher. Example power ratings include approximately 3.3 kW, 6.6 kW, and up to 14.4 kW.

Level 2 charging is faster than Level 1 charging because it provides more electrical power. It is commonly used in homes, workplaces, parking facilities, and public charging stations. As with Level 1 charging, the vehicle's onboard charger performs the AC-to-DC conversion, while the BMS coordinates battery charging limits.

DC CHARGING SYSTEMS

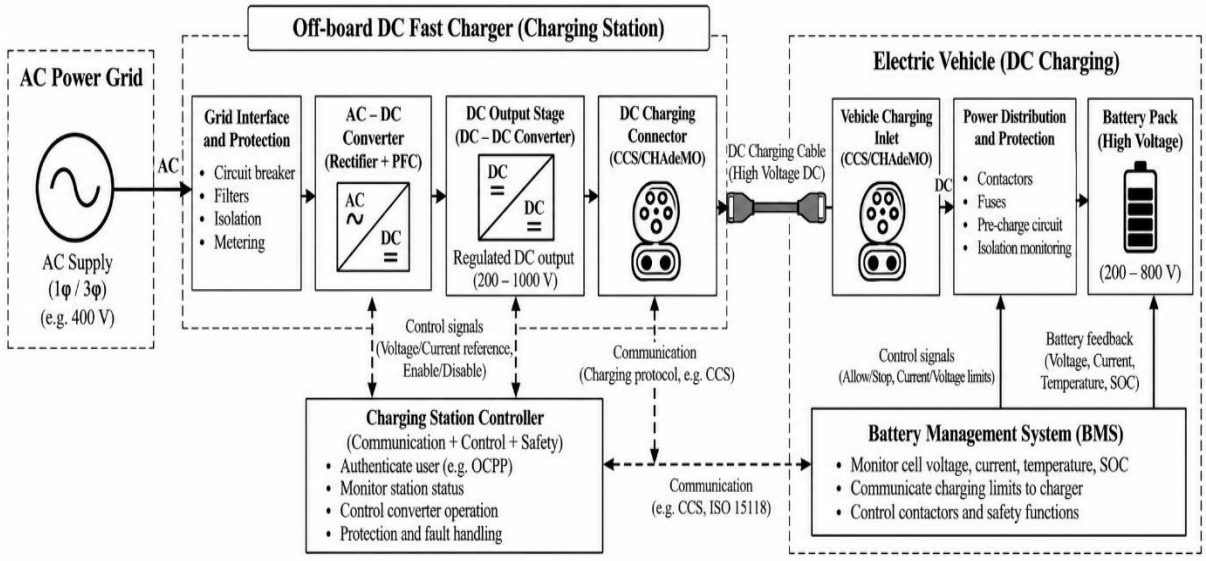
In a DC charging system, the main power-conversion equipment is installed outside the vehicle, usually at a dedicated charging station. The station receives AC power from the electrical grid and converts it into regulated DC. This DC power is then supplied directly to the vehicle's battery through the charging interface and associated protection and control systems.

Unlike AC charging, DC charging does not rely on the onboard charger to perform the main AC-to-DC conversion. The offboard charger contains the necessary power-electronic converters, control circuits, and protection systems. It communicates with the vehicle to determine the battery's voltage and appropriate charging limits. The charger then adjusts its output according to the vehicle's requirements and the limits communicated by the BMS.

DC chargers can deliver considerably more power than typical onboard AC chargers, allowing faster charging. They are commonly installed at public charging stations, highways, and fleet facilities. However, they require more complex external equipment, electrical infrastructure, protection, and, where necessary, cooling systems.

DC Charging Architecture

In a DC charging architecture, the power grid supplies AC electricity to an offboard charger. The charger converts the AC supply into regulated DC and delivers it directly to the vehicle's battery through a compatible charging interface. The BMS communicates battery requirements and charging limits to the charging system, and the charger adjusts its output accordingly.



DC Level 1 Charging

DC Level 1 charging uses a voltage range of 200–450 V DC, a maximum current of up to 80 A, and a power rating of up to 36 kW. The charging equipment is installed outside the vehicle and supplies regulated DC to the battery.

The actual charging time depends on battery capacity, temperature, state of charge, and the maximum charging current permitted by the vehicle.

DC Level 2 Charging

DC Level 2 charging uses a voltage range of 200–450 V DC, a maximum current of up to 200 A, and a power rating of up to 90 kW. The offboard charger regulates its output and communicates with the vehicle to ensure that the battery receives power within its permitted operating limits.

This charging level can substantially reduce charging time compared with lower-power charging systems. It requires suitable electrical connections, protection devices, and coordination between the charger and the BMS.

DC Level 3 Charging

DC Level 3 charging is specified uses a voltage range of 200–600 V DC, a maximum current of up to 400 A, and a power rating of up to 240 kW. It is intended for high-power charging applications where reducing the time required to recharge an EV is important.

The high power delivered by a DC Level 3 charger requires carefully designed converters, cables, connectors, protection, and thermal management. The charging station regulates its output according to the battery's voltage and current limits. Although high-power charging can reduce charging time considerably, the charging rate generally varies with battery state of charge, temperature, and charging capability.

Table 2 illustrates charge method electrical ratings according to SAE EV DC charging power levels.

Charge Method	Supplied DC Voltage Range	Maximum Current	Power Level
DC Level 1	200–450 V DC	≤80 A DC	≤36 kW
DC Level 2	200–450 V DC	≤200 A DC	≤90 kW
DC Level 3	200–600 V DC	≤400 A DC	≤240 kW

Comparison of AC and DC Charging Systems

The principal difference between AC and DC charging is the location of the AC-to-DC conversion stage. In AC charging, the conversion takes place inside the vehicle. In DC charging, the conversion takes place in the external charging station.

Parameter	AC charging	DC charging
Charger location	Onboard the vehicle	Offboard, at the charging station
Supply to vehicle	AC	Regulated DC
AC-to-DC conversion	Inside the vehicle	Outside the vehicle
Main power converter	Onboard charger	External charger
Typical charging power	Low to moderate	Moderate to very high
Charging speed	Generally slower	Generally faster
External infrastructure	Generally simpler	More complex and power-intensive
Common applications	Homes, workplaces, parking areas	Public fast-charging stations, highways, fleet facilities

AC charging systems are commonly installed onboard the vehicle, meaning that the charger is carried inside the vehicle. Since the charger operates in the vehicle environment, its components must meet automotive requirements. Reliability, thermal cycling, vibration, service life and warranty requirements are important design considerations. Onboard chargers can also increase vehicle cost, and manufacturers may face relatively low profit margins. AC Levels 1 and 2 are the dominant AC charging technologies in production vehicles.

CHARGING METHODS

Conductive Charging

Conductive charging is the most common charging method. It transfers electrical energy through a physical connection between the charging equipment and the vehicle using a cable, connector, and electrical contacts. Both AC and DC charging can use conductive connections. Suitable insulation, grounding, protection, and control arrangements are required to ensure safe energy transfer.

Inductive Charging

Inductive charging, also known as wireless charging, transfers energy through electromagnetic fields rather than a direct conductive connection. A transmitter coil in the charging pad produces a magnetic field, which induces voltage in a receiver coil installed in the vehicle. The received electrical energy is processed by power-electronic circuits to charge the battery. This method reduces the need to plug in a cable but requires suitable alignment, conversion equipment, and dedicated infrastructure.

Battery Swapping

Battery swapping is a method in which a depleted battery is removed from the vehicle and replaced with a fully charged battery at a swapping station. The depleted battery can then be recharged separately. This method reduces the time the vehicle must remain stationary for charging, provided that a compatible charged battery is available. However, it requires compatible battery packs, suitable mechanical arrangements, battery inventory, and dedicated infrastructure.

COMMON AC-BUS AND DC-BUS CHARGING STATIO ARCHITECTURES

Charging stations serving multiple EVs can be designed using a common AC bus or a common DC bus. The choice affects power conversion, energy distribution, efficiency, renewable-energy integration, and system complexity.

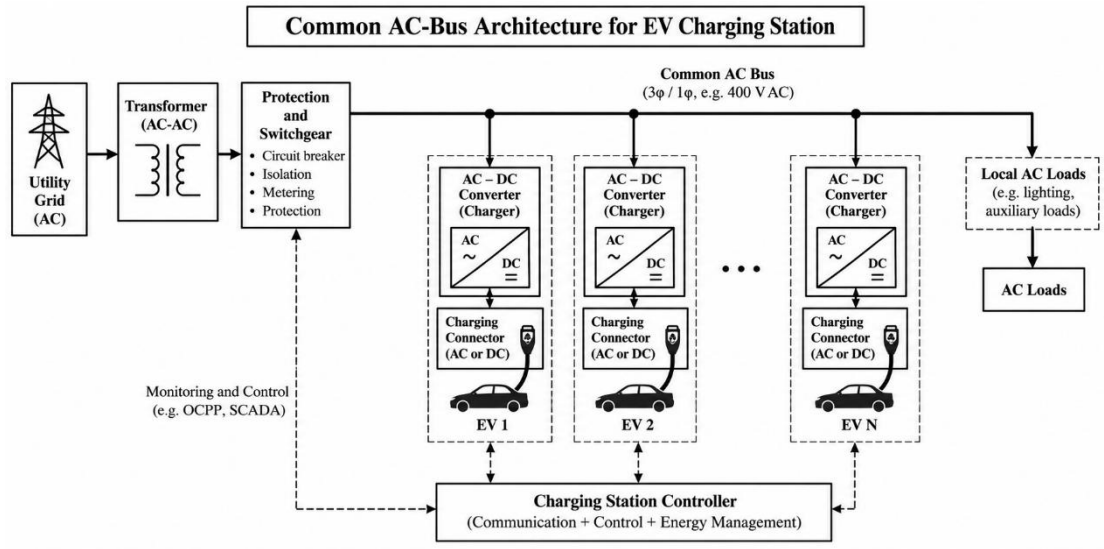
Common AC-Bus Architecture

In a common AC-bus architecture, the grid supplies AC power through the required transformer and distribution equipment to a shared AC bus. Each charging unit has its own power-conversion stage connected to this bus. In a station providing DC charging, each charging unit can contain its own AC–DC converter connected to the common AC bus.

This architecture uses established AC distribution technology and standard electrical components. It can be cost-effective, widely available, and relatively straightforward to protect. AC power can also be used directly by compatible local loads. However, when multiple individual converters are required, the total number of conversion stages increases system cost and complexity. High-power DC charging requires appropriate conversion equipment, and maintaining high efficiency and power quality can become more challenging as charging demand increases.

Advantages: Mature technology, standard components, relatively simple protection, compatibility with local AC loads, and good potential for stability and scalability.

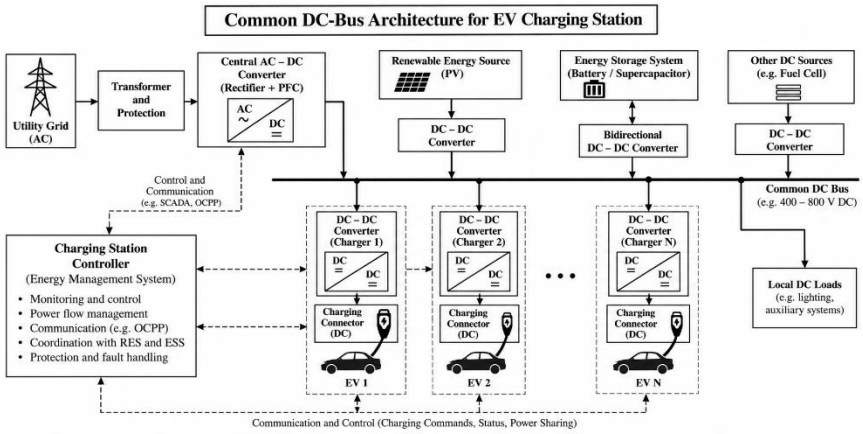
Disadvantages: Multiple converters can increase cost and losses. High-power charging may require additional conversion stages, while renewable-energy integration may require extra converters. Harmonics, power quality, and system stability must be managed appropriately.



Common DC-Bus Architecture

In a common DC-bus architecture, the grid-connected system converts AC power into DC through a central AC–DC converter, establishing a shared DC bus. Individual EV charging units can then use DC–DC converters to regulate the voltage and current supplied to each vehicle. Renewable energy sources and energy-storage systems may also be connected to the DC bus through suitable converters.

This arrangement can reduce the number of separate AC–DC conversion stages and provide an efficient interface for DC sources such as photovoltaic systems and battery energy storage. It can also offer flexible energy distribution among charging units. However, the central converter must be appropriately rated for the total power demand, and the system requires suitable DC protection, fault isolation, voltage regulation, and thermal management.



Advantages: Potentially high efficiency and power density, easier integration of compatible DC renewable sources and energy storage, flexible configuration, and improved coordination of multiple EV charging loads.

Disadvantages: Requires suitable DC-rated protection equipment, a central converter capable of handling the required power, and careful management of sudden load changes. The system can become more complex when multiple energy sources are connected, and high-power operation may require substantial cooling.

COMMUNICATION SYSTEM, BMS, AND GRID INTERFACE

The communication system enables the EV, charger, and other control units to exchange information needed for charging. This communication supports charging authorization, status monitoring, negotiation of charging limits, and control of the charging process. OCPP (Open Charge Point Protocol) is primarily used for communication between charging stations and their central management systems; it is distinct from the direct vehicle-to-charger charging-control protocols.

The Battery Management System (BMS) monitors the battery and helps ensure that charging remains within safe operating limits. It measures cell voltages, current, and temperature and estimates battery states such as SOC and SOH. The BMS communicates allowable charging limits and detects abnormal conditions that may require charging to be reduced or stopped. These functions help protect the battery and support efficient charging.

The grid interface connects the charging station to the electrical power network and provides the energy needed for charging. Depending on the installation, it may include transformers, switchgear, metering, protection devices, and power-conversion equipment. Smart-grid integration can support demand response and load balancing by coordinating charging demand with grid conditions. Renewable-energy sources and stationary energy-storage systems may also be integrated through suitable converters.